LESSON PLAN

PISTON ENGINE (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Stress & Strain, various types of stress and strain, explanation using stress strain diagram of various metals, Hooks law, factor of safety and other terminologies
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Stress & Strain, various types of stress and strain, explanation using stress strain diagram of various metals, Hooks law, factor of safety and other terminologies.
5. Specific Objectives	18.1.3.1 Trainee will be able to understand and know the heat treatments of Iron and alloys.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of tensile test principles using chart; Safety: No physical loading; Instructor supervision: Monitor diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the difference between stress and strain?
(b) Define Hooke's Law in your own words.
(c) Why is the Factor of Safety important in aircraft design?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Stress and Strain • Types of Stress: Tensile, Compressive, Shear • Types of Strain: Elastic, Plastic, Ultimate • Stress-Strain Diagram for Mild Steel • Stress-Strain Diagram for Brittle Materials	L & Ds	PPT, IB	15 min
(b)	• Hooke's Law and Proportionality Limit • Young's Modulus of Elasticity • Factor of Safety Definition • Terminologies: Yield Point, Breaking Load • Comparison of Ductile vs Brittle Behavior	L & Ds	PPT, PDF	20 min
(c)	• Application of Stress Concepts in Engines • Calculating Stress from Given Load • Determining Factor of Safety • Review of Key Terminologies • Summary of Stress-Strain Curves	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of the yield point on a stress-strain diagram.
(b) Calculate the factor of safety if the ultimate strength is 400 MPa and working stress is 100 MPa.
(c) Differentiate between elastic and plastic deformation with examples.

12. Home Assignment and Follow Up	Prepare a brief note on how understanding stress and strain aids in the selection of materials for piston engine components, linking to the next lesson on heat treatments.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Clarification required on specific heat treatment processes for Iron alloys mentioned in LP 02 to ensure continuity.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min